ค่าเฉลี่ย (s. mean) $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Sample Variance $\frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n(\bar{x})^2\right)$

Ex) 45 49 55 41 62

$\Sigma x_i^2 = (45^2 + 49^2 + 55^2 + 41^2 + 62^2)$

$= \frac{1}{4}(12976 - (5 \times (50.4)^2))$

$= 68.8$ #

ค่าเบี่ยงเบน (STD) $s = \sqrt{\text{Variane}}$

Ex) $\sqrt{68.8} = 8.2946$ #

Modifi (x + s. mean by c) $\bar{Y} = a\bar{X} + b$

· Variance $s^2_y = a^2 s^2_x$ · std $s_y = |a| s_x$

มัธยฐาน (s. Median) Odd = (n+1)/2

sort Even = หา $\bar{x}$ ของ n/2 กับ (n/2)+1

Ex) 41 45 49 55 62 705

∴ ((6/2) + ((6/2)+1))/2

= (49 + 55)/2 = 52 # ★ หาลำดับ

sort Quartile (if "int" else left/right)

$Q_{1,2,3} = 0.25(n+1), 0.50, 0.75$ ↳ s. Mean!

Percentile (if "int" else left/right)

Ex) n = 15 หา p = 20

∴ 0.20 (15+1) = 3.2 (use data 3, 4) find

(1640 + 2313)/2 = 1976.5 # s. Mean

ความหนาแน่น (Density) ★Histogram★

Relative Freq / Class Width

Ex) temp 56°–60°, RF = 0.05

กว้าง 5

∴ Density = 0.05 / 5 = 0.01 #

Boxplot (Quartile) กระจุก / กระจาย / outl.

Normal Distribution (Z-TABLE)

★ mean (μ) VAR (σ^2) std (σ)

$z = (X - \mu)/\sigma$

Ex) $\mu = 5$, $\sigma^2 = 9$

a) $P(X \leq 1) = \Phi((1-5)/3) = \Phi(-1.33)$

$= 1 - 0.9082 = 0.0918$ #

b) $P(X > 5.5) = 1 - \Phi((5.5-5)/3) = \Phi(0.17)$

$= 1 - 0.5675 = 0.4325$ #

c) $P(2 \leq X \leq 6) = \Phi(0.33) - (1 - \Phi(1))$

$= 0.6293 - (1 - 0.8413) = 0.4706$ #

CLT (โจทย์ให้ n มา) $(X - \bar{X})/(s/\sqrt{n})$

Ex) $\bar{X} = 22$ $s = 10$ $n = 60$, $P(X < 23)$?

$P(x < (23-22)/(10/\sqrt{40}))$

$= \Phi(0.63) = 0.7357$ #

ช่วงความเชื่อมั่น (C.I.) $\bar{X} \pm z_{a/2} \cdot (s/\sqrt{n})$

Ex) n = 88 $\bar{x}$ = 250 s = 15, 95 %

a = 1 − 0.95 = 0.05 a/2 = 0.025

$z_{0.025}$ = 1 − 0.025 = 1.96 ↳ 0.975 (z-table)

250 ± 1.96 · (15/$\sqrt{88}$) = (− , +)

Ex) n = 66 $\bar{X}$ = 116 s = 12, (108.26, 121.34)

① 121.34 = 116 + X · (12/$\sqrt{66}$) ★ย้ายข้าง

② X ≈ 3.385 $z_{a/2}$ = 3.39 a/2 = 0.9997

③ หา a → 1 − a/2 = 0.9997 = 0.9994 ★ย้ายข้าง 99.94 %

Hypothesis $z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

Ex) n = 4 $\bar{x}$ = 50 s = 4 μ = 51

t-table d.f = n − 1 = 3

t = (50 − 51)/(4/$\sqrt{4}$) = −1/2 = −0.5

P-Value = 0.25 < P-Value < 0.40

Regression (r = 0 → Uncorrelated)

$r = (1/(n-1)) \sum_{i=1}^{n} ((x_i - \bar{x})/s_x)((y_i - \bar{y})/s_y)$

$s_{x,y} = \sqrt{\sum_{i=1}^{n} (x_i - \bar{x})/(n-1)}$

สมการถดถอย

$y = \hat{B}_0 + (\hat{B}_1) x$

$\hat{B}_0 = \bar{y} - \hat{B}_1 \bar{x}$

$\hat{B}_1 = \sum_{i=0}^{n} ((x_i - \bar{x})(y_i - \bar{y})) / \sum_{i=0}^{n} (x_i - \bar{x})^2$

Example zone !

PROBABILITY [$\bar{x}$ = 22 σ = 10]

1) 20 < X < 23

Sol^n X = (20 − 22)/10 = 1 − Φ(0.2)

= 1 − (1 − 0.5793) = 0.5793

X = (23 − 22)/10 = Φ(0.1)

= 0.5398

∴ 0.5398 − 0.5793 = 0.0395 #

2) X > 19

Sol^n X = (19 − 22)/10 = 1 − Φ(0.3)

= 1 − (1 − 0.6179) = 0.6179 #

3) Percentile 60 (→ 0.6000) ↳ z-table

Sol^n $\bar{X}$ + z-score · σ = 22 + 0.25 · 10

= 24.5 #

4) X < 23 n = 40

Sol^n (23 − 22)/(10/$\sqrt{40}$) = Φ(0.63)

= 0.7357 #

C. Interval [n = 90 $\bar{x}$ = 252 σ = 17]

SE = $\sigma/\sqrt{n}$ → 17/$\sqrt{90}$ = 1.792

1) (249.3701, 254.6502)

Sol^n 252 − 249.3701 = 2.6299

→ 2.6299 / 1.792 (SE) = Φ(1.47)

= 0.9292 → 92.92 %

→ 100 − 92.92 = 7.08

→ 92.92 − 7.08 = 85.84 % #

2) 85 %

↳ 0.925 z-table 1.44

Sol^n 100 − 85 = (15/2) + 85 = 92.5

∴ 252 ± (1.44 · 1.792 (SE)) 2.5805

= (249.4195, 254.5805) #

Hypothesis [$\bar{x}$ = 98 s = 6 n = 6]

H0 : u >= 99 H1 : u < 99

1) test statistic

Sol^n z = (98 − 99)/(6/$\sqrt{6}$) = −0.4082 ↳ z-tb

∴ P-value = 1 − 0.6591 = 0.3409 #

2) 7 % (0.07) = ไม่สามารถปฏิเสธ

3) 3 % (0.03) = "

★ ถ้า P-Value > ระดับนัยสำคัญ = ไม่ปฏิเสธ

Note!

z-score neg(−)

↳ PCT < 50

Ex) 25 (0.2500)

→ 0.7500 ↳ z-table

$\bar{X}$ + (−z · s)

C.I. Level (2 ข้าง)

68 %	=	1
90 %	=	1.645
95 %	=	1.96
99 %	=	2.58
99.7 %	=	3

ข้างเดียว ↓

90 %	=	1.28
95 %	=	1.645
99 %	=	2.33

★ ข้อมูล < 30 → t-table

$\bar{X} \pm t_{n-1/a/2} \cdot (s/\sqrt{n})$

Setup ON → SHIFT → 9 (Reset) → 1 (Setup) → = → AC

Cal Menu → 2 (Graph) → $y = a + bx$

→ ใส่เลข (ใช้ลูกศรขยับ) → = → AC

มีปุ่ม OPTN

↳ กด OPTN → 3 (Regress)

ไม่มี

↳ กด shift → 1 → Regress